

Digital Control System Analysis and Design

Chap. 1 Introduction

Chap. 2 Discrete Time Systems and the z-Transform

Chap. 3 Sampling and Reconstruction

Chap. 4 Open-Loop Discrete-Time Systems

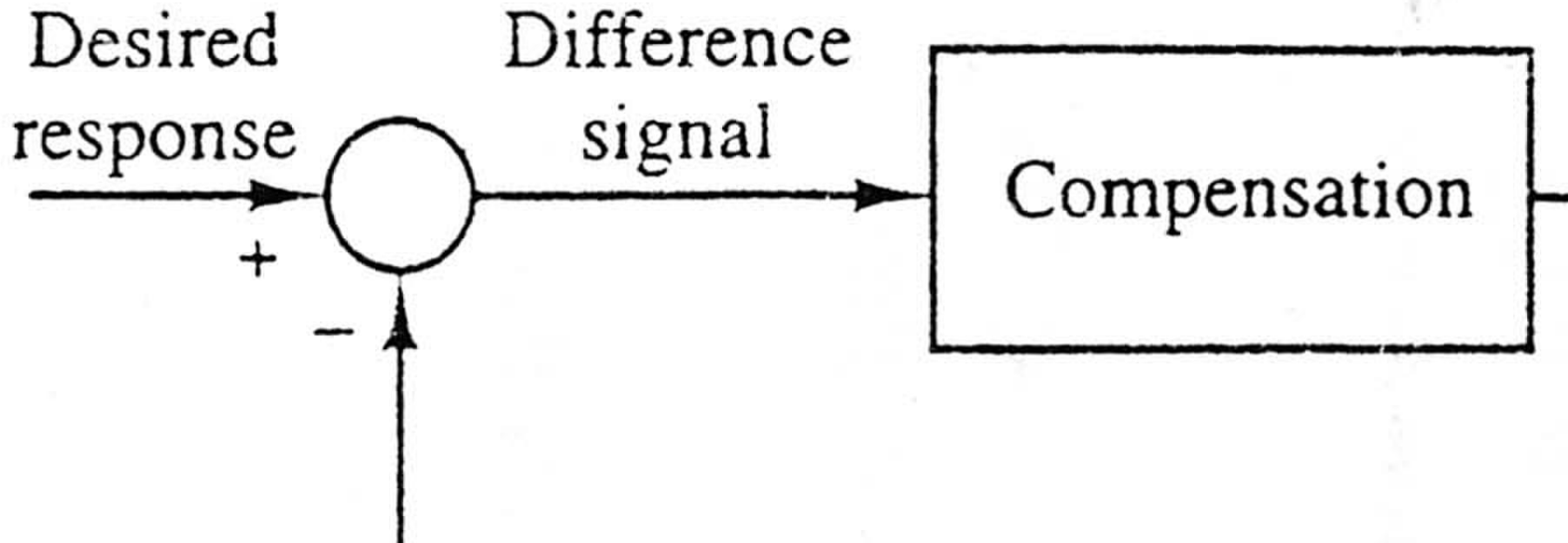
Chap. 5 Closed-Loop Systems

Chap. 6 System Time-Response Characteristics

Chap. 7 Stability Analysis Techniques

Chap. 8 Digital Controller Design

1.1 Overview



- Compensator, controller, filter
- Linear time-invariant discrete-time systems

1.2 Digital Control System

Control System: 4 Areas to be developed

- a) Process or Plant -- Dynamic systems**
- b) Measurement Technology --- Sensors**
- c) Computer Technology ----- VLSI, CPU, I/O**
- d) Control Theory ----- Discrete and continuous control**
 - . Sampled Theorem**
 - . Difference Eqn.**
 - . z-Transform**
 - . State-Space Model**

- . System Identification**
- . Adaptive Control**
- . Optimal Control**
- . Neural Fuzzy Control**
- . Stochastic Processes**

Advantages of digital Control

- 1) Digital components are **less susceptible to aging** and environmental variations
- 2) **Less sensitive** to noise and disturbance
- 3) Digital processors are more compact and lightweight
- 4) **Single chip** (e.g. DSP) can be made very versatile and powerful
- 5) **Cost down**
- 6) More **flexibility in programming**
- 7) Changing a design doesn't require an alternation of hardware
- 8) More reliable
- 9) Provide **improved sensitivity** to parameter variations

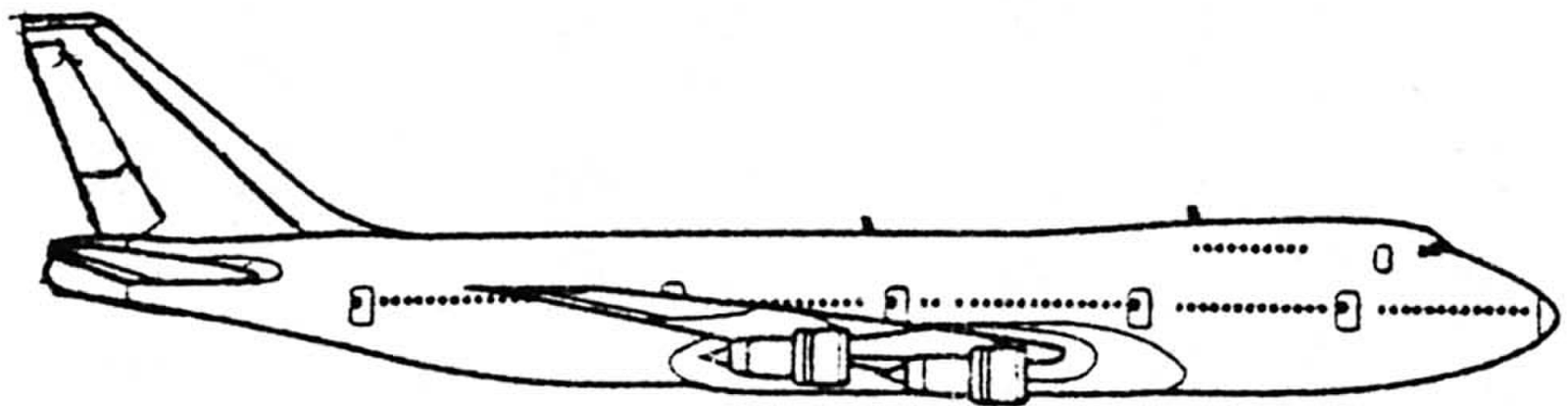
Disadvantages

1) Computing speed limit.

Analog controllers --- Real Time and Resolution is theoretically infinite

2) The finite wavelength of digital processor often translate into system **instability** in limit cycle

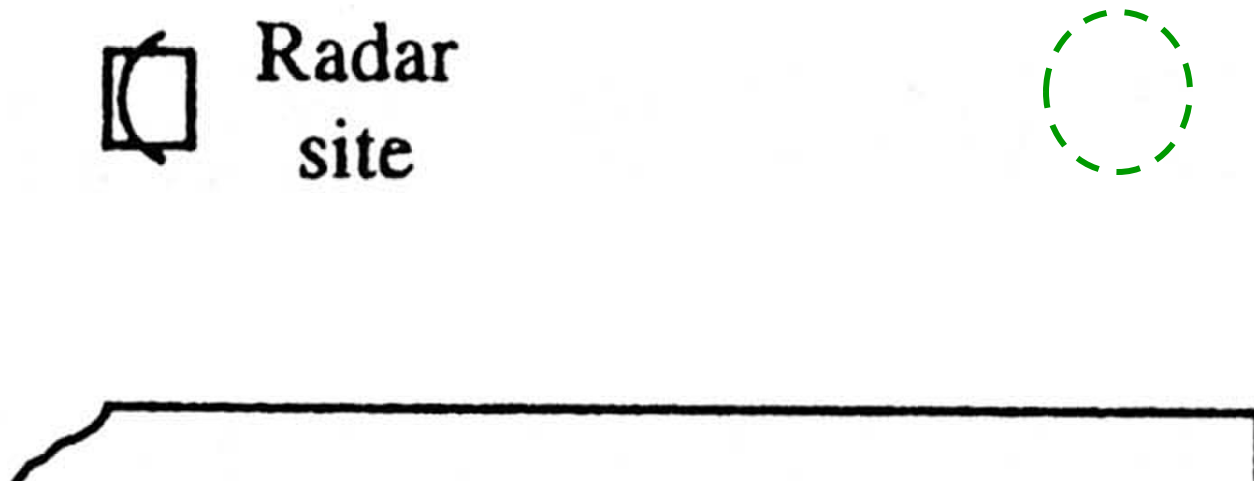
3) Computing speed cost **time delays** in control System---may be unstable reason in closed loop system



Aircraft

傾斜角

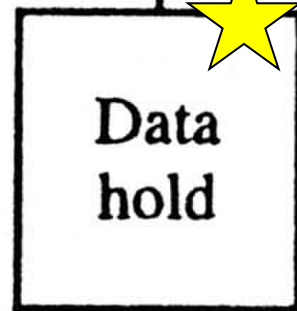
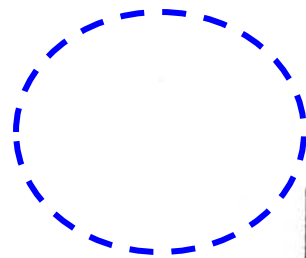
仰角



- Lateral control: maintain $y(t)$ small

wind disturbance
→

reconstruction



$\phi(t)$
Bank command



$w(t)$

sensor noise

No sensor is perfect!

1.3 The Control Problem

What are the Control System Main Problems?

- **Disturbance Rejection:** 閉迴路
- **Steady-State Errors:** 準
- **Transient Response (incl. Stability) :** 快、穩
- **Sensitivity of Parameter Variation:** : 狠

打靶: 快、穩、狠、準!

Solve Control Problem

1. **Choosing sensors**
2. **Choosing actuators**
3. **Develop plant, sensor & actuator models**
4. **Design a controller** (Based on control law and model)
5. **Evaluating the Design by Simulation & Testing**
6. **Iterating the Procedure**

數學

將實際問題→數學化

真實世界



Problem formulation



將數學解答→實現化

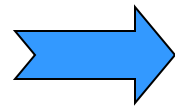
Physical
system

1.4 Satellite Model

1.5 Servomotor System Model

1.6 Temperature Control System

- See Summary (P18)
- Problems: 1-2, 1-3, 1-4, 1-5, 1-6



Test Bank